

Final Year B. Chem. Engg. (Sem. VIII) – Continuous Assessment Test 1

Wednesday, 7th February 2024, Time: 1 h, Maximum Marks: 10

CET1515 – Innovations in Chemical Engineering & Technology

Attempt any two questions.

- Q1. Typically, a chemical engineer fulfils a variety of job functions (such as Manufacturing, Engineering, R&D, Technical Support etc. among others) at their company in the Chemical Process Industry (CPI). Discuss innovation opportunities for chemical engineers with specific examples of how to go about innovating.
- Q2. Ammonia manufacturing has evolved during the past several decades with several commercial technologies from BASF, KBR, Haldor-Topsoe, ThyssenKrupp and Linde. Discuss technological advancements in the ammonia industry by listing the improvements in ammonia process technology.
- Q3. Discuss the phased gate process for innovation projects with a suitable diagram.

-----X-----

Final Year B. Chem. Engg. (Sem. VIII) – Continuous Assessment Test 1

Wednesday, 7th February 2024, Time: 1 h, Maximum Marks: 10

CET1515 – Innovations in Chemical Engineering & Technology

Attempt any two questions.

- Q1. Which questions should be raised when conducting a success modes and effects analysis for innovation projects? Explain with suitable examples.
- Q2. Methanol synthesis processes from ICI, Kellogg and Lurgi are well-established. Discuss innovations in methanol production during the past decades, especially related to improvements in technology, reactors, catalysts, feedstock, and scaleup.
- Q3. Discuss the technological framework for process innovation with a suitable diagram.

-----X-----

Final Year B.Chem. Engg. (Sem. VIII) – Continuous Assessment Test 1 (Repeat)

Wednesday, 29th February 2024, Time: 1 h, Maximum Marks: 10

CET1515 – Innovations in Chemical Engineering & Technology

Attempt any two questions.

- Q1. Discuss how innovation is similar to a basic chemical engineering process. List the steps taken to generate value for customers and within your organization.
- Q2. Explain how innovation in the chemical industry can contribute to success in the business.
- Q3. Discuss innovative chemical engineering solutions in ENERGY SECTOR or FOOD & AGRICULTURE.

-----X-----

Final Year B.Chem. Engg. (Sem. VIII) – Mid-Semester Examination

Thursday, 14th March 2024, Time: 1.30 – 2.30 pm, Maximum Marks: 15

CET1515 – Innovations in Chemical Engineering & Technology

Attempt any three questions.

- Q1. How do we evaluate the maturity or readiness of a particular technology? Explain using examples.
- Q2. Discuss examples of innovations in distillation technology using suitable diagrams.
- Q3. How does Peter Drucker define innovation? Discuss features of the four types of innovation (incremental, architectural, radical and disruptive). Discuss disruptive innovation in detail.
- Q4. Discuss innovation opportunities for chemical engineers at their company in Manufacturing, Engineering, R&D and Technical Support with specific examples of how to go about innovating in the Chemical Process Industry (CPI).

-----X-----

Final Year B.Chem. Engg. (Sem. VIII) April-May 2024 Examination

Friday, 3rd May 2024, Time: 2–4 pm, Maximum Marks: 25

CET1515 – Innovations in Chemical Engineering & Technology

Attempt any five questions.

- Q1. Which are the characteristics of today's and tomorrow's chemical sectors? Explain the role of innovation in designing for a green chemistry future with a suitable example.
- Q2. As engineers, sometimes we just assume that to make money off an invention, you have to file a patent application. Explain why that's simply not true. When does patenting an invention make sense and when does keeping it a trade secret make sense?
- Q3. Discuss innovative solutions adopted by the fertilizer industry for the production of sustainable fertilizers.
- Q4. Discuss the four main categories of innovation in the chemical industry, viz. product, manufacturing process, business process, and business and operating model innovation. Explain how these different categories are leveraged to boost the business during the various phases of the product life cycle.
- Q5. Discuss examples of innovative chemical engineering solutions in sustainability, food and healthcare.